Machine Learning

14-04-26

$$x_1, x_2 \longrightarrow \boxed{\text{log regr}} \longrightarrow y \in [0,1]$$

$$y = P(y = +ve \mid x_1, x_2)$$

$$\log\left(\frac{p}{1-p}\right) = w_1 x_1 + w_2 x_2 + b$$

this is how input features are related to probability

lets put $z = \underbrace{w_1 x_1 + w_2 x_2 + b}$

eqn of the linear decision boundary

$$\log\left(\frac{p}{1-p}\right) = z$$

$t = \log(u)$

$e^t = u$

$$\frac{p}{1-p} = e^z$$

$$p = (1-p)e^z$$

$$p = e^z - pe^z$$

$$p + pe^z = e^z$$

$$p(1+e^z) = e^z$$

$$p = \frac{e^z}{1+e^z}$$

$$p = \frac{e^z/e^z}{(1+e^z)/e^z}$$

$$p = \frac{e^z/e^z}{(1+e^z)/e^z}$$

$$p = \frac{1}{1/e^z + 1}$$

$$\boxed{p = \frac{1}{1+e^{-z}}}$$ → sigmoid function

$$y = \sigma(x)$$

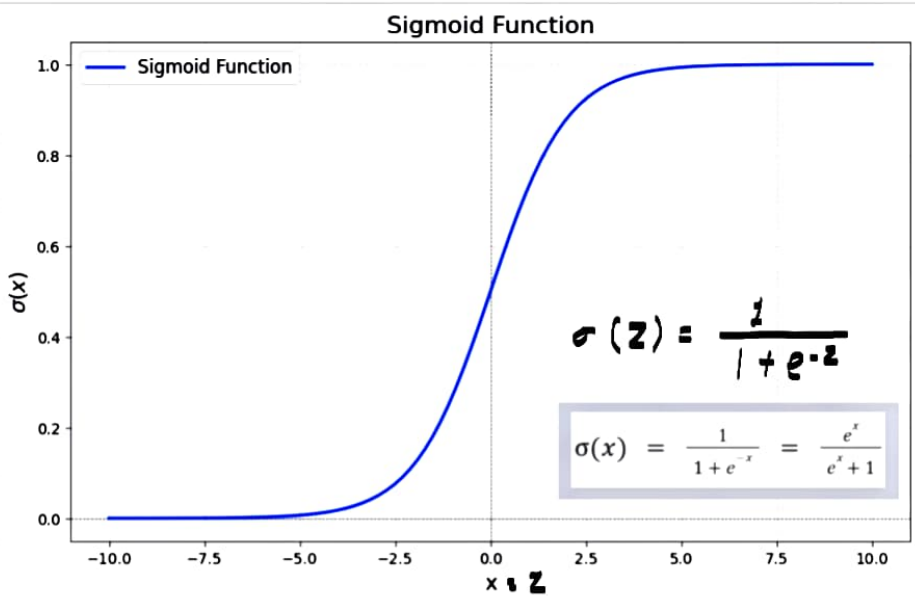
$$y = \frac{1}{1+e^{-x}}$$

$$\boxed{p = \sigma(z)}$$

where $z = w_1 x_1 + w_2 x_2 + b$

$$P(y=1 \mid x_1, x_2) = \sigma(z) = \sigma(w_1 x_1 + w_2 x_2 + b)$$

$$\left[P(Y=1 \mid x_1, x_2) = \frac{1}{1 + e^{-(w_1 x_1 + w_2 x_2 + b)}} \right]$$



- converts any input into an output lying betn 0 to 1

Hypothesis function :

$$P(Y=1 \mid x_1, x_2) = \sigma(w_1 x_1 + w_2 x_2 + b)$$

	x_1	x_2	Y_{act}	Y_{pred}
d_1	1	0.1	1	1
d_2	4	0.5	0	0
d_3	6	0.8	1	1
	⋮	⋮	⋮	

If $Y_{act} = 1$ then we want model to predict $Y_{pred} = 1$

for $Y_{pred} = 1$ / +ve

$$P(Y=1 \mid x_1, x_2) \approx 1$$
$$P(Y=0 \mid x_1, x_2) \approx 0$$

$$\sigma(w_1 x_1 + w_2 x_2 + b) \approx 1$$

- training objective for datapoint d_1 will be finding optimal values of w_1 w_2 and b such that
$$\sigma(w_1 x_1 + w_2 x_2 + b) \approx 1$$

for d_2 we want $Y_{pred} = 0$ / -ve

$$P(Y=1 \mid x_1, x_2) \approx 0$$
$$P(Y=0 \mid x_1, x_2) \approx 1$$

$$\sigma(w_1 x_1 + w_2 x_2 + b) \approx 0$$

- During model training, we need to find optimal values of w_1 w_2 and b such that if $Y_{act} = 1$ then $\sigma(z) \approx 1$ and
if $Y_{act} = 0$ then $\sigma(z) \approx 0$

Loss Function

if $Y_{act} = 1$ then $\sigma(z) \approx 1$ i.e. maximize $\sigma(z)$

if $Y_{act} = 0$ then $\sigma(z) \approx 0$ ie minimize $\sigma(z)$

minimize $P(Y=1 \mid x_1, x_2)$

$P(Y=1 \mid x_1, x_2) = 1 - P(Y=0 \mid x_1, x_2)$

minimize $P(Y=1 \mid x_1, x_2) \Rightarrow$ maximize $P(Y=0 \mid x_1, x_2)$

$\Rightarrow$ maximize $[1 - P(Y=1 \mid x_1, x_2)]$

$\Rightarrow$ maximize $1 - \sigma(z)$

if $Y_{act} = 1$ then maximize $\sigma(z)$

if $Y_{act} = 2$ then maximize $1 - \sigma(z)$

put both objectives in single eqn

objective : maximize $\left\{ [\sigma(z)]^{Y_{act}} \times [1 - \sigma(z)]^{1 - Y_{act}} \right\}$

when $Y_{act} = 1 \Rightarrow$ maximize $\left\{ [\sigma(z)]^{1} \times [1 - \sigma(z)]^{1-1} \right\}$

$\Rightarrow \max \{ \sigma(z) \}$

when $Y_{act} = 0 \Rightarrow \max \left\{ [\sigma(z)]^{0} \times [1 - \sigma(z)]^{1-0} \right\}$

$\Rightarrow \max \{ 1 - \sigma(z) \}$

for every train datapoint having x_1 x_2 Y_{act}

$$\max\left\{ [\sigma(z)]^{Y_{act}} \cdot [1-\sigma(z)]^{1-Y_{act}} \right\}$$

$$\max\left\{ P(Y=1|x_1,x_2)^{Y_{act}} \cdot [1-P(Y=1|x_1,x_2)]^{1-Y_{act}} \right\}$$

lets put $p = p(Y=1|x_1,x_2)$

objective : $\max\left\{ p^{Y_{act}} \cdot (1-p)^{1-Y_{act}} \right\}$

$$\max\left\{ p^{Y} \cdot (1-p)^{1-Y} \right\}$$

	x_1	x_2	Y
d_1			
d_2			
d_3			
$\vdots$			
d_n			

for d_i train datapoint

$$\max\left\{ p_i^{Y_i} \times (1-p_i)^{1-Y_i} \right\}$$

where $p_i = \sigma(Z_i)$

$$Z_i = w_1 x_{i1} + w_2 x_{i2} + b$$

$$Y_i = Y_{act} \text{ for } d_i$$

for total 'n' datapoints

$\max[p_1^{Y_1}(1-p_1)^{1-Y_1}]$ and $\max[p_2^{Y_2} \cdot (1-p_2)^{1-Y_2}]$ and ... $\max[p_n^{Y_n} \cdot (1-p_n)^{1-Y_n}]$

for all 'n' train datapoints, objective is

$$\max[p_1^{y_1}.(1-p_1)^{1-y_1}] \text{ and } \max[p_2^{y_2}.(1-p_2)^{1-y_2}] \ldots \text{ and } \max[p_n^{y_n}.(1-p_n)^{1-y_n}]$$

as above is joint probability of 'n' probabilities,

objective :

$$\max[p_1^{y_1}.(1-p_1)^{1-y_1}] \cap \max[p_2^{y_2}.(1-p_2)^{1-y_2}] \ldots \cap \max[p_n^{y_n}.(1-p_n)^{1-y_n}]$$

eg.

	height	weight	class
$d_1 \rightarrow$	20	60	A
$d_2 \rightarrow$	30	50	A
$d_3 \rightarrow$	40	45	B

all train datapoints are independent of each other in this case

lets consider each datapoint as an event

d_1 = event A $\quad$ d_2 = event B

when A and B are independent of each other

$$\boxed{P(A \cap B) = P(A).P(B)}$$

objective :

$$\max[p_1^{y_1}.(1-p_1)^{1-y_1}] \cap \max[p_2^{y_2}.(1-p_2)^{1-y_2}] \ldots \cap \max[p_n^{y_n}.(1-p_n)^{1-y_n}]$$

as all datapoints are independent of each other

objective :

$$\max\{p_1^{y_1}.(1-p_1)^{1-y_1} \times p_2^{y_2}.(1-p_2)^{1-y_2} \times p_3^{y_3}.(1-p_3)^{1-y_3} \ldots \times p_n^{y_n}.(1-p_n)^{1-y_n}\}$$

objective:

$$\max \left\{ p_1^{y_1} \cdot (1-p_1)^{1-y_1} \times p_2^{y_2} \cdot (1-p_2)^{1-y_2} \times p_3^{y_3} \cdot (1-p_3)^{1-y_3} \cdots \times p_n^{y_n} \cdot (1-p_n)^{1-y_n} \right\}$$

As above is multiplication of multiple probability values (which are small values betn 0 to 1) we face numerical instability issue

$p_1 = 0.1 \quad p_2 = 0.8 \quad p_3 = 0.2 \quad p_4 = 0.1 \quad p_5 = 0.3$

$p_1 \times p_2 \times p_3 \times p_4 \times p_5 = 0.00048$

when decimal values become really small, they can't be stored in computers accurately. this is numerical instability issue.

solution: Apply log function

after applying log function, new objective becomes

$$\max \left\{ \log \left[p_1^{y_1} \cdot (1-p_1)^{1-y_1} \times p_2^{y_2} \cdot (1-p_2)^{1-y_2} \cdots \times p_n^{y_n} \cdot (1-p_n)^{1-y_n} \right] \right\}$$

$$\log(t \times u) = \log t + \log u$$

$$\max \left\{ \log \left[p_1^{y_1} \cdot (1-p_1)^{1-y_1} \right] + \log \left[p_2^{y_2} \cdot (1-p_2)^{1-y_2} \right] + \cdots \log \left[p_n^{y_n} \cdot (1-p_n)^{1-y_n} \right] \right\}$$

$$\boxed{\max \sum_{i=1}^{n} \log \left[p_i^{y_i} (1-p_i)^{1-y_i} \right]}$$

objective:

$$\max \sum_{i=1}^{n} \log\left[p_i^{y_i} \cdot (1-p_i)^{1-y_i} \right]$$

$$\max \sum_{i=1}^{n} \log(p_i^{y_i}) + \log\left[(1-p_i)^{1-y_i}\right]$$

$$\log(t^u) = u \log(t)$$

$$\max \sum_{i=1}^{n} y_i \log(p_i) + (1-y_i)\log(1-p_i)$$

$$\Downarrow$$

$$\min -\sum_{i=1}^{n} y_i \log(p_i) + (1-y_i)\log(1-p_i)$$

Loss function:

$$\boxed{L = -\sum_{i=1}^{n} y_i \log(p_i) + (1-y_i)\log(1-p_i)}$$

this is binary cross entropy loss function

BCE loss is used for binary classification model

Objective of logistic regression model training is to minimize BCE loss.

model 1

$Y_1 = 1$

$P_1 = \sigma(Z_1) = 0.6$

$Y_{act} \not\approx Y_{pred}$

model 2

$Y_1 = 1$

$P_1 = \sigma(Z_1) = 0.9$

$Y_{act} \approx Y_{pred}$

here model 2 looks better than model 1

BCE loss for model 2 < BCE loss for model 1

$$\text{loss1} = -\sum_{i=1}^{n} Y_i \log(P_i) + (1-Y_i)\log(1-P_i)$$

$$= -[1 \times \log(0.6) + (1-1)\log(1-0.6)]$$

$$= -\log(0.6)$$

$$= 0.221$$

$$\text{loss 2} = -\sum_{i=1}^{n} Y_i \log(P_i) + (1-Y_i)\log(1-P_i)$$

$$= -[1 \times \log(0.9) + (1-1)\log(1-0.9)]$$

$$= -\log(0.9)$$

$$= 0.045$$

when $Y_{pred} \approx Y_{act}$, BCE loss is less

lesser the $\underbrace{\text{BCE loss}}_{\text{gradient descent}} \Rightarrow$ better the model